

II BE Computer Engineering  
Class Test-III Nov 2023  
3ACRC1 Applied Mathematics-III

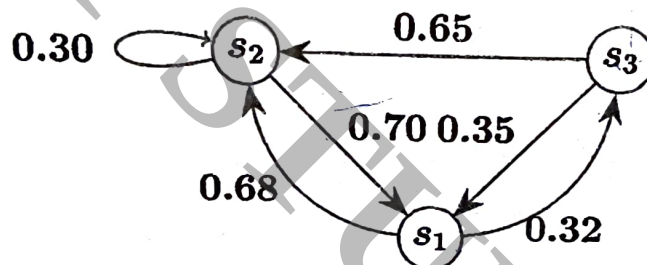
Note-Attempt all the questions.  
Time-70 mins.

Maximum Marks [20]

Q.1. If  $X(t) = A \cos(\omega t + \phi)$ , where  $A$  and  $\phi$  are independent random variables and  $\phi$  is uniformly distributed in  $(-\pi, \pi)$ , then show that the random process  $\{X(t)\}$  is a *WSS* process.

[5]

Q.2. A Markov chain is given by the following transition diagram-



Show that  $F_{33} = 1$  and hence the state  $s_3$  is recurrent.

[5]

Q.3. Two different types of drugs A and B were tried on certain patients for increasing weight. Five persons were given drug A and 7 persons were given the drug B. The increase in weight (in kgs) is given below-

Drug A 3.6 5.5 5.9 4.1 1.4

Drug B 4.5 3.6 5.5 6.8 2.7 3.6 5.0

Can it be concluded at  $\alpha = 0.05$  that the drug B is more effective than drug A in increasing weight?

[5]

Q.4. The following sample data pertains to the shipment received by a large firm from four different vendors:

| Quality                  | Vendors |    |    |    |
|--------------------------|---------|----|----|----|
|                          | A       | B  | C  | D  |
| Rejected                 | 15      | 19 | 13 | 12 |
| Imperfect but acceptable | 15      | 19 | 13 | 12 |
| Perfect                  | 15      | 19 | 13 | 12 |

Test at  $\alpha = 0.05$ , whether the three vendors ship products of equal quality.

[5]

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